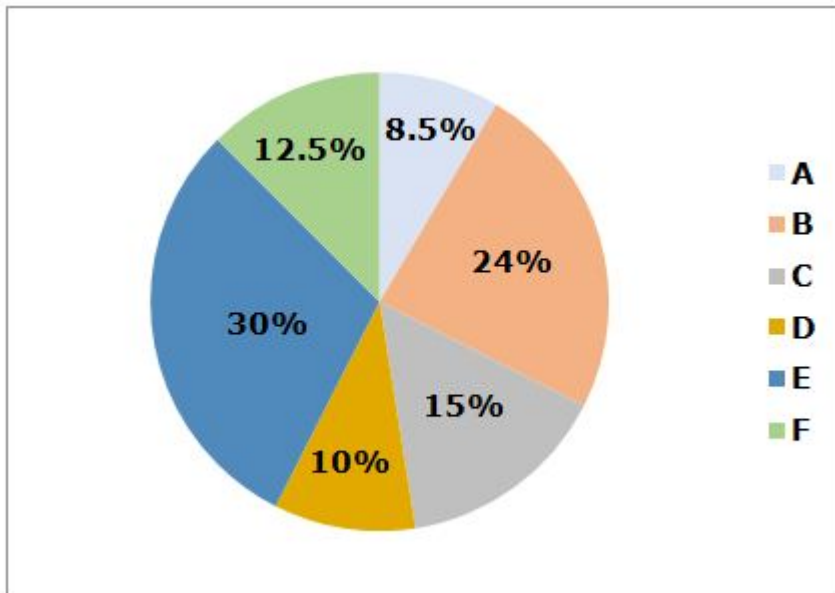


1. Questions

Study the following data carefully and answer the questions:

Data given below is related to the number of photos taken by six photographers A, B, C, D, E and F of some tourists at a tourist place.

Pie chart given below shows the percentage distribution of the number of photos taken by each of the given photographers.



Notes:

- 1:** Out of the total photos taken by a photographer, some photos were clear and the remaining photos were blurred.
- 2:** The ratio of the number of clear photos to the number of blurred photos taken by **Photographer A** is 4: 3.
- 3:** The ratio of the number of clear photos to the number of blurred photos taken by **Photographer E** is 16: 5.
- 4:** The number of clear photos taken by **Photographer E** is 252 more than that taken by **Photographer A**.

If 25% of the total number of photos taken by Photographer B, are blurred photos and the number of clear photos taken by Photographer B is 110% more than that taken by Photographer D, then find the average number of blurred photos taken by Photographer B, D and E?

- a. 72
- b. 68
- c. 76
- d. 64
- e. 70

2. Questions

Photographer A charges ₹X per photo (only for clear photos), photographer E charges ₹(X – Y) per photo (only for clear photos) and the total amount received by Photographer A and B together is ₹3580. If photographer A had charged ₹(X + Y + 3) per photo (only for clear photos), then his earning would have been increased by what percentage?

- a. $66\frac{2}{3}\%$
- b. 75%
- c. 60%
- d. 83.5%
- e. Can't be determined

3. Questions

If the difference between the number of blurred photos taken by Photographer A and C is 9, then the number of blurred photos taken by Photographer C will be either be ____% or ____% of the number of clear photos taken by Photographer C.

Find that which of the following values can be filled in the blanks in any order?

- a. 25, $33\frac{1}{3}$
- b. 40, 20
- c. 20, $33\frac{1}{3}$
- d. 25, 40
- e. None of these

4. Questions

If the ratio of the number of clear photos to the number of blurred photos taken by Photographer C is 25: 17, if the total number of photos taken by Photographer G is 90% of that taken by Photographer C and the number of clear photos taken by Photographer G is double of the number of blurred photos taken by Photographer C, then find the average of the number of clear photos taken by Photographer C and the number of blurred photos taken by Photographer G?

- a. 78
- b. 84
- c. 72
- d. 80
- e. 64

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5. Questions

Photographer F charges Rs. P per photo. If all the photos taken by Photographer F were clear, he would have got a total of Rs. 3150. But it was found that some of the photos taken by him were

blurred, due to which he got Rs. 990 less. Find the ratio of the number of clear photos to the number of blurred photos taken by Photographer F?

- a. $(P + 6): (P - 7)$
- b. $(P + 4): (P - 13)$
- c. $(P + 7): (P - 4)$
- d. $(P + 5): (P - 11)$
- e. $(P + 6): (P - 13)$

6. Questions

Study the following data carefully and answer the questions:

The data given below is related to the number of questions attempted by a student in five different exams A, B, C, D and E and the maximum number of questions in each exam is different.

Table given below shows the following data:

Exam	Total number of questions	Number of questions attempted as % of the total number of questions	The number of questions answered right as % of the number of questions attempted
A	$(X + Y)$	64%	75%
B	$(5Y - 10)$	55%	$66\frac{2}{3}\%$
C	$2(X - Y)$	62%	$83\frac{1}{3}\%$
D	160	70%	50%
E	X	76%	87.5%

Notes:

- 1: The difference between the number of questions answered right in exams B and D is 32.
- 2: The number of questions attempted in exam B is not a perfect square.
- 3: The number of questions attempted in exam E is 20 more than that attempted in exam B.

If in exam B, 1 mark is given for every right answered question and Y/X mark is deducted for every wrong answered question and in exam D, 1 mark is given for every right answered question and $Y/(X - Y)$ mark is deducted for every wrong answered question then what will be the ratio of the total marks obtained in exam B to that in exam D?

- a. 77: 56
- b. 3: 4
- c. 11: 7

d. 231: 112

e. 154: 111

7. Questions

If in exams A, C and E, 2 marks are given for every right answered question and 0.5 marks are deducted for every wrong answered question, then find that which of the following pair of (Exam name – Marks obtained) is/are not correct?

a. (Exam A – 220)

b. (Exam C – 284.5)

c. (Exam E – 256.5)

d. Both (a) and (b)

e. Both (a) and (c)

8. Questions

If the maximum number of questions attempted in an exam is $P\%$ more than the 2nd maximum number of questions attempted in an exam and the minimum number of questions attempted in an exam is $Q\%$ of the 2nd maximum number of questions attempted in an exam, then find the ratio of P to Q?

a. 13: 35

b. 39: 70

c. 13: 56

d. None of these

e. 39: 35

9. Questions

If the total number of questions in exam F is $[(X - Y)/4]\%$ more than that in exam D and the number of questions attempted in exam F is 90% of the total number of questions in exam F, out of which $18(2/11)\%$ questions are answered wrong, then find that the number of questions answered right in exam F is what per cent more than the number of questions answered right in exam A?

a. $[X - 3Y - 20]\%$

b. $[(X/5) - 5]\%$

c. $[Y - 10]\%$

d. $[(X - Y)/4]\%$

e. None of these

10. Questions

In which exam, the number of right answered questions as % of the total number of questions is 2nd highest?

- a. A
- b. D
- c. C
- d. B
- e. E

11. Questions

Study the following data carefully and answer the questions:

Data given below is related to the rates of interest in four schemes A, B, C and D.

If an amount of Rs. 8000 is invested in scheme A, the difference between the compound interest and simple interest after 2 years is Rs. X. If an amount of Rs. 4000 is invested in scheme B, the difference between the compound interest and simple interest after 2 years is Rs. (X + 17.5). The ratio of the rate of interest in scheme A to that in scheme B is 3: 5 and the rate of interest in scheme A is 93.75% of that in scheme C. If an amount of Rs. 7500 is invested in scheme D, then the simple interest received from scheme D after 2 years will be Rs. 300 less than that of received from scheme A after 2 years.

A man invested Rs. 100X in scheme C for 't' years, Rs. (100X + 500) in scheme D for (t + 1) years and Rs. (100X – 500) in scheme B for (t + 2) years. If the simple interest received from scheme D is Rs. 180 more than that received from scheme C, then find the simple interest received from scheme B?

- a. Rs. 1720
- b. Rs. 1800
- c. Rs. 2000
- d. Rs. 3000
- e. Rs. 1880

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12. Questions

If the rate of interest in scheme B was 28% less than the original rate of interest, then find that which of the following statements would be true?

I: The ratio of the rate of interest in scheme A to that in scheme B would be 5: 6.

II: The simple interest received from scheme B on Rs. 3500 after 2 years would be Rs. 630.

III: The new rate of interest in scheme B would be a prime number.

- a. Only II and III

- b. Only II
- c. Only I and III
- d. Only III
- e. Only I and II

13. Questions

If the simple interest received from scheme C on Rs. (P + 500) after 20 months (the interest is calculated annually) is Rs. 1400, then find that the simple interest received from scheme D on Rs. (P – 2500) after 15 months (the interest is calculated annually) will lie between which of the following ranges?

- a. 540 and 560
- b. 548 and 568
- c. Both (a) and (b)
- d. 555 and 575
- e. Both (b) and (d)

14. Questions

Compute the approximate value of x and answer the following questions.

$$19.99^2 - 12.99^2 + (4832.05 - x) / 19.88 = 500$$

- a. - 538
- b. - 528
- c. - 548
- d. - 500
- e. - 558

15. Questions

$$\sqrt{1295.88 \times 49.90 + 1079.60 \div 24.05 - 44.05 \times 7.99} = x$$

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- a. 1493
- b. 1483
- c. 1393
- d. 1593
- e. 1563

16. Questions

$$\sqrt{2208.99 \times 12.05 + 59.98 - 31.52\% \text{ of } 449.99} = x$$

- a. 450
- b. 480
- c. 420
- d. 460
- e. 400

17. Questions

$$660\sqrt{x} = (23.99 \times 14.98) - 227.98$$

- a. $1/27$
- b. $1/9$
- c. $1/16$
- d. $1/25$
- e. $1/49$

18. Questions

Each of the questions below consists of a question and three statements I, II and III are given below it. You have to decide whether the data provided in the statements are sufficient to answer the question. Read all the statements and give answer.

The lengths of trains A and B are 235 m and 350 m respectively and they cross each other in 15 seconds, while running in opposite direction. Find the time taken by train A to cross a 'd' m long platform?

Statement I: Train B crosses a $(d - 130)$ m long tunnel in 25 seconds.

Statement II: Time taken by train A to cross a man in train B is 47 seconds, while running in same direction.

Statement III: Time taken by train A to cross a $(d + 110)$ m long bridge is 25 seconds.

- a. Statement I alone is sufficient to answer the question.
- b. Statements I and III together are sufficient to answer the question.
- c. Statement II alone is sufficient to answer the question.
- d. Both (a) and (C)
- e. Both (b) and (c)

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19. Questions

Given that $(M + 3) \times (N + 2) \times (O - 1) = 385$ (M, N, and $O > 2$), & the value of $M + N - O$ lies between 2 and 5?

Statement I: M is the largest integer.

Statement II: Integer values of M and O are equal.

Statement III: Sum of integer values of N and O is a prime number and M^O is a perfect square integer.

P: Statement II alone is sufficient.

Q: Statement III alone is sufficient.

R: Statements I and III together are sufficient.

- a. Either P or Q
- b. Either P or R
- c. Only R
- d. Only P
- e. Only Q

20. Questions

Monthly salary of a person is Rs. 30000 if remains present for full month. In a particular month of 30 days, he took 'x' half pay leaves and 'y' no pay leaves, find the total deduction in his salary due to half pay leaves only.

Half pay leaves mean half salary for that day and no pay leaves mean no salary for that day.

Statement I: He received total Rs. 25500 as salary for that month.

Statement II: Deduction in salary due to half pay leaves is Rs. 500 more than the deduction in salary due to no pay leaves.

Statement III: Total deduction in his salary due to both half pay leaves and no pay leaves together is Rs. 4500.

P: Both the statements I and II together are sufficient.

Q: Both the statements I and III together are sufficient.

R: Both the statements II and III together are sufficient.

- a. Either P or Q
- b. Only Q
- c. Only R
- d. Either P or R
- e. All P, Q, and R

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Explanations:

1. Questions

Let the numbers of photos taken by **Photographer** A, B, C, D, E and F are $8.5x$, $24x$, $15x$, $10x$, $30x$ and $12.5x$ respectively.

So, the number of clear photos taken by **Photographer** A = $8.5x * (4/7) = (34x/7)$

And the number of clear photos taken by **Photographer** E = $30x * (16/21) = (160x/7)$

Since, the number of clear photos taken by **Photographer** E is 252 more than that taken by **Photographer** A.

So,

$$\Rightarrow (160x/7) - (34x/7) = 252$$

$$\Rightarrow 126x/7 = 252$$

$$\Rightarrow x = 14$$

Now, the total number of photos taken by **Photographer** A = $8.5 * 14 = 119$

The total number of photos taken by **Photographer** B = $24 * 14 = 336$

The total number of photos taken by **Photographer** C = $15 * 14 = 210$

The total number of photos taken by **Photographer** D = $10 * 14 = 140$

The total number of photos taken by **Photographer** E = $30 * 14 = 420$

And the total number of photos taken by **Photographer** F = $12.5 * 14 = 175$

Answer: B

Since, the total number of photos taken by **Photographer** B = 336

So, the number of blurred photos taken by **Photographer** B = 25% of 336 = 84

And the number of clear photos taken by **Photographer** B = $336 - 84 = 252$

Since, the number of clear photos taken by **Photographer** B is 110% more than that taken by **Photographer** D.

So, the number of clear photos taken by **Photographer** D = $252 * (100/210) = 120$

Since, the total number of photos taken by **Photographer** D = 140

So, the number of blurred photos taken by **Photographer** D = $140 - 120 = 20$

Since, the total number of photos taken by **Photographer** E = 420

And the ratio of the number of clear photos to the number of blurred photos taken by **Photographer** E is 16: 5.

So, the number of blurred photos taken by **Photographer** E = $420 * (5/21) = 100$

And the required average = $(84 + 20 + 100)/3 = 68$

2. Questions

Let the numbers of photos taken by **Photographer** A, B, C, D, E and F are $8.5x$, $24x$, $15x$, $10x$, $30x$ and $12.5x$ respectively.

So, the number of clear photos taken by **Photographer** A = $8.5x * (4/7) = (34x/7)$

And the number of clear photos taken by **Photographer** E = $30x * (16/21) = (160x/7)$

Since, the number of clear photos taken by **Photographer** E is 252 more than that taken by **Photographer** A.

So,

$$\Rightarrow (160x/7) - (34x/7) = 252$$

$$\Rightarrow 126x/7 = 252$$

$$\Rightarrow x = 14$$

Now, the total number of photos taken by **Photographer** A = $8.5 * 14 = 119$

The total number of photos taken by **Photographer** B = $24 * 14 = 336$

The total number of photos taken by **Photographer** C = $15 * 14 = 210$

The total number of photos taken by **Photographer** D = $10 * 14 = 140$

The total number of photos taken by **Photographer** E = $30 * 14 = 420$

And the total number of photos taken by **Photographer** F = $12.5 * 14 = 175$

Answer: E

Since, the ratio of the number of clear photos to the number of blurred photos taken by **Photographer** A = 4: 3

So, the number of clear photos taken by **Photographer** A = $119 * (4/7) = 68$

Since, the ratio of the number of clear photos to the number of blurred photos taken by **Photographer** E = 16: 5

So, the number of clear photos taken by **Photographer** E = $420 * (16/21) = 320$

Since, the total amount received by **Photographer** A and E together = ₹3580

So,

$$\Rightarrow 68 * X + 320 * (X - Y) = 3580$$

$$\Rightarrow 68X + 320X - 320Y = 3580$$

$$\Rightarrow 388X - 320Y = 3580$$

$$\Rightarrow 97X - 80Y = 895$$

Since, the values of 'X' and 'Y' can't be determined.

So, the required answer can't be determined.

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3. Questions

Let the numbers of photos taken by **Photographer** A, B, C, D, E and F are $8.5x$, $24x$, $15x$, $10x$, $30x$ and $12.5x$ respectively.

So, the number of clear photos taken by **Photographer** A = $8.5x * (4/7) = (34x/7)$

And the number of clear photos taken by **Photographer** E = $30x * (16/21) = (160x/7)$

Since, the number of clear photos taken by **Photographer** E is 252 more than that taken by **Photographer** A.

So,

$$\Rightarrow (160x/7) - (34x/7) = 252$$

$$\Rightarrow 126x/7 = 252$$

$$\Rightarrow x = 14$$

Now, the total number of photos taken by **Photographer** A = $8.5 * 14 = 119$

The total number of photos taken by **Photographer** B = $24 * 14 = 336$

The total number of photos taken by **Photographer** C = $15 * 14 = 210$

The total number of photos taken by **Photographer** D = $10 * 14 = 140$

The total number of photos taken by **Photographer** E = $30 * 14 = 420$

And the total number of photos taken by **Photographer** F = $12.5 * 14 = 175$

Answer: D

Since, the total number of photos taken by **Photographer** A = 119

So, the number of blurred photos taken by **Photographer** A = $119 * (3/7) = 51$

Since, the difference between the number of blurred photos taken by **Photographer** A and C = 9

So, the number of blurred photos taken by **Photographer** C will either be $(51 + 9) = 60$ or $(51 - 9) = 42$

Case I: When the number of blurred photos taken by **Photographer** C = 60

Since, the total number of photos taken by **Photographer** C = 210

And the number of clear photos taken by **Photographer** C = $210 - 60 = 150$

So, the required percentage = $(60/150) * 100 = 40\%$

Case II: When the number of blurred photos taken by **Photographer** C = 42

Since, the total number of photos taken by **Photographer** C = 210

And the number of clear photos taken by **Photographer** C = $210 - 42 = 168$

So, the required percentage = $(42/168) * 100 = 25\%$

So, the number of blurred photos taken by **Photographer** C will either be 25% or 40% of the number of clear photos taken by **Photographer** C.

4. Questions

Let the numbers of photos taken by **Photographer** A, B, C, D, E and F are $8.5x$, $24x$, $15x$, $10x$, $30x$ and $12.5x$ respectively.

So, the number of clear photos taken by **Photographer** A = $8.5x * (4/7) = (34x/7)$

And the number of clear photos taken by **Photographer** E = $30x * (16/21) = (160x/7)$

Since, the number of clear photos taken by **Photographer** E is 252 more than that taken by **Photographer** A.

So,

$$\Rightarrow (160x/7) - (34x/7) = 252$$

$$\Rightarrow 126x/7 = 252$$

$$\Rightarrow x = 14$$

Now, the total number of photos taken by **Photographer** A = $8.5 * 14 = 119$

The total number of photos taken by **Photographer** B = $24 * 14 = 336$

The total number of photos taken by **Photographer** C = $15 * 14 = 210$

The total number of photos taken by **Photographer** D = $10 * 14 = 140$

The total number of photos taken by **Photographer** E = $30 * 14 = 420$

And the total number of photos taken by **Photographer** F = $12.5 * 14 = 175$

Answer: C

Since, the total number of photos taken by **Photographer** C = 210

So, the number of clear photos taken by **Photographer** C = $210 * (25/42) = 125$

And the number of blurred photos taken by **Photographer** C = $210 - 125 = 85$

Since, the total number of photos taken by **Photographer** G = 90% of 210 = 189

And the number of clear photos taken by **Photographer** G = $2 * 85 = 170$

So, the number of blurred photos taken by **Photographer** G = $189 - 170 = 19$

Required average = $(125 + 19)/2 = 72$

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5. Questions

Let the numbers of photos taken by **Photographer** A, B, C, D, E and F are $8.5x$, $24x$, $15x$, $10x$, $30x$ and $12.5x$ respectively.

So, the number of clear photos taken by **Photographer** A = $8.5x * (4/7) = (34x/7)$

And the number of clear photos taken by **Photographer** E = $30x * (16/21) = (160x/7)$

Since, the number of clear photos taken by **Photographer** E is 252 more than that taken by **Photographer** A.

So,

$$\Rightarrow (160x/7) - (34x/7) = 252$$

$$\Rightarrow 126x/7 = 252$$

$$\Rightarrow x = 14$$

Now, the total number of photos taken by **Photographer A** = $8.5 * 14 = 119$

The total number of photos taken by **Photographer B** = $24 * 14 = 336$

The total number of photos taken by **Photographer C** = $15 * 14 = 210$

The total number of photos taken by **Photographer D** = $10 * 14 = 140$

The total number of photos taken by **Photographer E** = $30 * 14 = 420$

And the total number of photos taken by **Photographer F** = $12.5 * 14 = 175$

Answer: A

Since, the total number of photos taken by F = 175

So,

$$\Rightarrow P = 3150/175$$

$$\Rightarrow P = 18$$

Since, he got Rs. 990 less.

So, the number of blurred photos taken by **Photographer F** = $990/18 = 55$

And the number of clear photos taken by **Photographer F** = $175 - 55 = 120$

Required ratio = $120: 55 = 24: 11 = (P + 6): (P - 7)$

6. Questions

Since, the number of questions answered right in exam D = 50% of 70% of $160 = 56$

So, the number of questions answered right in exam B will either be $56 + 32 = 88$ or $56 - 32 = 24$

Case I: If the number of questions answered right in exam B = 88

So, the number of questions attempted in exam B = $88 * (300/200) = 132$

Case II: If the number of questions answered right in exam B = 24

So, the number of questions attempted in exam B = $24 * (300/200) = 36$

Since, the number of questions attempted in exam B is not a perfect square.

So, case II can be neglected.

And the number of questions attempted in exam B = 132

Since, the total number of questions in exam B = $132 * (100/55) = 240$

So,

$$\Rightarrow (5Y - 10) = 240$$

$$\Rightarrow Y = (240 + 10)/5$$

$$\Rightarrow Y = 50$$

Since, the number of questions attempted in exam E is 20 more than that attempted in exam B.

So, the number of questions attempted in exam E = $132 + 20 = 152$

And the total number of questions in exam E = $152 * (100/76) = 200$

So, X = 200

Exam	Total no. of questions	No. of questions attempted	No. of questions answered right
A	$200 + 50 = 250$	$64\% \text{ of } 250 = 160$	$75\% \text{ of } 160 = 120$
B	$5 * 50 - 10 = 240$	$55\% \text{ of } 240 = 132$	$66(2/3)\% \text{ of } 132 = 88$
C	$2 * (200 - 50) = 300$	$62\% \text{ of } 300 = 186$	$83(1/3)\% \text{ of } 186 = 155$
D	160	$70\% \text{ of } 160 = 112$	$50\% \text{ of } 112 = 56$
E	200	$76\% \text{ of } 200 = 152$	$87.5\% \text{ of } 152 = 133$

Answer: D

In exam B:

The number of questions answered right = 88

The number of questions answered wrong = $132 - 88 = 44$

Since, the negative marking = $Y/X = 50/200 = 1/4$ marks

So, the total marks obtained in exam B:

$$= 88 * 1 - 44 * (1/4)$$

$$= 88 - 11$$

$$= 77$$

In exam D:

The number of questions answered right = 56

The number of questions answered wrong = $112 - 56 = 56$

Since, the negative marking is of $Y/(X - Y) = 50/(200 - 50) = 1/3$ marks

So, the total marks obtained in exam D:

$$= 56 * 1 - 56 * (1/3)$$

$$= (168 - 56)/3$$

$$= 112/3$$

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Required ratio = 77: 112/3 = 231: 112

7. Questions

Since, the number of questions answered right in exam D = 50% of 70% of 160 = 56

So, the number of questions answered right in exam B will either be $56 + 32 = 88$ or $56 - 32 = 24$

Case I: If the number of questions answered right in exam B = 88

So, the number of questions attempted in exam B = $88 * (300/200) = 132$

Case II: If the number of questions answered right in exam B = 24

So, the number of questions attempted in exam B = $24 * (300/200) = 36$

Since, the number of questions attempted in exam B is not a perfect square.

So, case II can be neglected.

And the number of questions attempted in exam B = 132

Since, the total number of questions in exam B = $132 * (100/55) = 240$

So,

$$\Rightarrow (5Y - 10) = 240$$

$$\Rightarrow Y = (240 + 10)/5$$

$$\Rightarrow Y = 50$$

Since, the number of questions attempted in exam E is 20 more than that attempted in exam B.

So, the number of questions attempted in exam E = $132 + 20 = 152$

And the total number of questions in exam E = $152 * (100/76) = 200$

So, X = 200

Exam	Total no. of questions	No. of questions attempted	No. of questions answered right
A	$200 + 50 = 250$	$64\% \text{ of } 250 = 160$	$75\% \text{ of } 160 = 120$
B	$5 * 50 - 10 = 240$	$55\% \text{ of } 240 = 132$	$66(2/3)\% \text{ of } 132 = 88$
C	$2 * (200 - 50) = 300$	$62\% \text{ of } 300 = 186$	$83(1/3)\% \text{ of } 186 = 155$
D	160	$70\% \text{ of } 160 = 112$	$50\% \text{ of } 112 = 56$
E	200	$76\% \text{ of } 200 = 152$	$87.5\% \text{ of } 152 = 133$

Answer: B

In exam A:

The number of right answered question = 120

The number of wrong answered question = $160 - 120 = 40$

So, the marks obtained = $(120 * 2) - (40 * 0.5) = 220$

In exam C:

The number of right answered question = 155

The number of wrong answered question = $186 - 155 = 31$

So, the marks obtained = $(155 * 2) - (31 * 0.5) = 294.5$

In exam E:

The number of right answered question = 133

The number of wrong answered question = $152 - 133 = 19$

So, the marks obtained = $(133 * 2) - (19 * 0.5) = 256.5$

The correct pairs are = (Exam A – 220), (Exam C – 294.5), (Exam E – 256.5)

8. Questions

Since, the number of questions answered right in exam D = 50% of 70% of 160 = 56

So, the number of questions answered right in exam B will either be $56 + 32 = 88$ or $56 - 32 = 24$

Case I: If the number of questions answered right in exam B = 88

So, the number of questions attempted in exam B = $88 * (300/200) = 132$

Case II: If the number of questions answered right in exam B = 24

So, the number of questions attempted in exam B = $24 * (300/200) = 36$

Since, the number of questions attempted in exam B is not a perfect square.

So, case II can be neglected.

And the number of questions attempted in exam B = 132

Since, the total number of questions in exam B = $132 * (100/55) = 240$

So,

$$\Rightarrow (5Y - 10) = 240$$

$$\Rightarrow Y = (240 + 10)/5$$

$$\Rightarrow Y = 50$$

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Since, the number of questions attempted in exam E is 20 more than that attempted in exam B.

So, the number of questions attempted in exam E = $132 + 20 = 152$

And the total number of questions in exam E = $152 * (100/76) = 200$

So, X = 200

Exam	Total no. of questions	No. of questions attempted	No. of questions answered right
A	$200 + 50 = 250$	$64\% \text{ of } 250 = 160$	$75\% \text{ of } 160 = 120$
B	$5 * 50 - 10 = 240$	$55\% \text{ of } 240 = 132$	$66(2/3)\% \text{ of } 132 = 88$
C	$2 * (200 - 50) = 300$	$62\% \text{ of } 300 = 186$	$83(1/3)\% \text{ of } 186 = 155$
D	160	$70\% \text{ of } 160 = 112$	$50\% \text{ of } 112 = 56$
E	200	$76\% \text{ of } 200 = 152$	$87.5\% \text{ of } 152 = 133$

Answer: C

Since, the maximum number of questions attempted in an exam = 186

And the 2nd maximum number of questions attempted in an exam = 160

So,

$$\Rightarrow P = [(186 - 160)/160] * 100$$

$$\Rightarrow P = 16.25\%$$

Since, the minimum number of questions attempted in an exam = 112

And the 2nd maximum number of questions attempted in an exam = 160

So,

$$\Rightarrow Q = (112/160) * 100$$

$$\Rightarrow Q = 70\%$$

Required ratio = P: Q = 16.25: 70 = 13: 56

9. Questions

Since, the number of questions answered right in exam D = 50% of 70% of 160 = 56

So, the number of questions answered right in exam B will either be $56 + 32 = 88$ or $56 - 32 = 24$

Case I: If the number of questions answered right in exam B = 88

So, the number of questions attempted in exam B = $88 * (300/200) = 132$

Case II: If the number of questions answered right in exam B = 24

So, the number of questions attempted in exam B = $24 * (300/200) = 36$

Since, the number of questions attempted in exam B is not a perfect square.

So, case II can be neglected.

And the number of questions attempted in exam B = 132

Since, the total number of questions in exam B = $132 * (100/55) = 240$

So,

$$\Rightarrow (5Y - 10) = 240$$

$$\Rightarrow Y = (240 + 10)/5$$

$$\Rightarrow Y = 50$$

Since, the number of questions attempted in exam E is 20 more than that attempted in exam B.

So, the number of questions attempted in exam E = $132 + 20 = 152$

And the total number of questions in exam E = $152 * (100/76) = 200$

So, $X = 200$

Exam	Total no. of questions	No. of questions attempted	No. of questions answered right
A	$200 + 50 = 250$	$64\% \text{ of } 250 = 160$	$75\% \text{ of } 160 = 120$
B	$5 * 50 - 10 = 240$	$55\% \text{ of } 240 = 132$	$66(2/3)\% \text{ of } 132 = 88$
C	$2 * (200 - 50) = 300$	$62\% \text{ of } 300 = 186$	$83(1/3)\% \text{ of } 186 = 155$
D	160	$70\% \text{ of } 160 = 112$	$50\% \text{ of } 112 = 56$
E	200	$76\% \text{ of } 200 = 152$	$87.5\% \text{ of } 152 = 133$

Answer: B

Since, the total number of questions in exam D = 160

So, the total number of questions in exam F = $[100 + (X - Y)/4]\% \text{ of } 160 = [100 + (200 - 50)/4]\% \text{ of } 160$
 $= [100 + 37.5]\% \text{ of } 160 = 220$

The number of questions attempted in exam F = $90\% \text{ of } 220 = 198$

The number of questions answered wrong in exam F = $18(2/11)\% \text{ of } 198 = 36$

So, the number of questions answered right in exam F = $198 - 36 = 162$

Since, the number of questions answered right in exam A = 120

So, the required percentage = $[(162 - 120)/120] * 100 = 35\%$

From option (a):

Since, the value of $(X - 3Y - 20) = (200 - 3 * 50 - 20) = 30$

So, option (a) is not correct.

From option (b):

Since, the value of $[(X/5) - 5] = [(200/5) - 5] = 35$

So, option (b) is correct.

From option (c):

Since, the value of $(Y - 10) = (50 - 10) = 40$

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So, option (c) is not correct.

From option (d):

Since, the value of $[(X - Y)/4] = [(200 - 50)/4] = 37.5$

So, option (d) is not correct.

Hence, option (b) is correct.

10. Questions

Since, the number of questions answered right in exam D = 50% of 70% of 160 = 56

So, the number of questions answered right in exam B will either be $56 + 32 = 88$ or $56 - 32 = 24$

Case I: If the number of questions answered right in exam B = 88

So, the number of questions attempted in exam B = $88 * (300/200) = 132$

Case II: If the number of questions answered right in exam B = 24

So, the number of questions attempted in exam B = $24 * (300/200) = 36$

Since, the number of questions attempted in exam B is not a perfect square.

So, case II can be neglected.

And the number of questions attempted in exam B = 132

Since, the total number of questions in exam B = $132 * (100/55) = 240$

So,

$$\Rightarrow (5Y - 10) = 240$$

$$\Rightarrow Y = (240 + 10)/5$$

$$\Rightarrow Y = 50$$

Since, the number of questions attempted in exam E is 20 more than that attempted in exam B.

So, the number of questions attempted in exam E = $132 + 20 = 152$

And the total number of questions in exam E = $152 * (100/76) = 200$

So, X = 200

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Exam	Total no. of questions	No. of questions attempted	No. of questions answered right
A	$200 + 50 = 250$	$64\% \text{ of } 250 = 160$	$75\% \text{ of } 160 = 120$
B	$5 * 50 - 10 = 240$	$55\% \text{ of } 240 = 132$	$66(2/3)\% \text{ of } 132 = 88$
C	$2 * (200 - 50) = 300$	$62\% \text{ of } 300 = 186$	$83(1/3)\% \text{ of } 186 = 155$
D	160	$70\% \text{ of } 160 = 112$	$50\% \text{ of } 112 = 56$
E	200	$76\% \text{ of } 200 = 152$	$87.5\% \text{ of } 152 = 133$

Answer: C

The number of right answered question as % of the total number of questions in exam A = $(120/250) * 100 = 48\%$

The number of right answered question as % of the total number of questions in exam B = $(88/240) * 100 = 36.67\%$

The number of right answered question as % of the total number of questions in exam C = $(155/300) * 100 = 51.67\%$

The number of right answered question as % of the total number of questions in exam D = $(56/160) * 100 = 35\%$

The number of right answered question as % of the total number of questions in exam E = $(133/200) * 100 = 66.5\%$

So, in exam C, the number of right answered questions as % of the total number of questions, is 2nd highest.

11. Questions

Since, the ratio of the rate of interest in scheme A to that in scheme B = 3: 5

So, let the rates of interest in schemes A and B be '3r%' and '5r%' respectively.

Since, the difference between compound interest and simple interest on Rs. 8000, which is invested in scheme A, after 2 years is Rs. X.

So,

$$\Rightarrow 8000 * (3r/100)^2 = X \text{ -----(1)}$$

Since, the difference between compound interest and simple interest on Rs. 4000, which is invested in scheme B, after 2 years is Rs. (X + 17.5).

So,

$$\Rightarrow 4000 * (5r/100)^2 = (X + 17.5) \text{ -----(2)}$$

By equation (2) – (1):

$$\Rightarrow 4000 * (5r/100)^2 - 8000 * (3r/100)^2 = X + 17.5 - X$$

$$\Rightarrow 4000 * (25r^2/10000) - 8000 * (9r^2/10000) = 17.5$$

$$\Rightarrow 4 * (25r^2/10) - 8 * (9r^2/10) = 17.5$$

$$\Rightarrow 100r^2 - 72r^2 = 175$$

$$\Rightarrow r^2 = 6.25$$

$$\Rightarrow r = 2.5$$

The rate of interest in scheme A = $3 * 2.5 = 7.5\%$

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The rate of interest in scheme B = $5 * 2.5 = 12.5\%$

From equation (1):

$$8000 * (3r/100)^2 = X$$

$$8000 * (7.5/100)^2 = X$$

$$X = 45$$

Since, the rate of interest in scheme A is 93.75% of that in scheme C.

$$\text{So, the rate of interest in scheme C} = 7.5 * (100/93.75) = 8\%$$

If an amount of Rs. 7500 is invested in scheme D, then the simple interest received from scheme D after 2 years will be Rs. 300 less than that of received from scheme A after 2 years.

Let the rate of interest in scheme D be R%.

$$\text{Since, the simple interest received from scheme A after 2 years} = (8000 * 7.5 * 2)/100 = \text{Rs. 1200}$$

So, the simple interest received by from scheme D after 2 years:

$$\Rightarrow (7500 * R * 2)/100 = 1200 - 300$$

$$\Rightarrow 150R = 900$$

$$\Rightarrow R = 6$$

So, the rate of interest in scheme D = 6%

Answer: C

$$\text{The amount invested in scheme C} = 100 * 45 = \text{Rs. 4500}$$

$$\text{The amount invested in scheme D} = 4500 + 500 = \text{Rs. 5000}$$

$$\text{The amount invested in scheme B} = 4500 - 500 = \text{Rs. 4000}$$

Since, the SI received from scheme D is Rs. 180 more than that received from scheme C.

So,

$$[5000 * 6 * (t + 1)]/100 - [4500 * 8 * t]/100 = 180$$

$$300t + 300 - 360t = 180$$

$$t = 2$$

Now, the SI received from scheme B:

$$\Rightarrow [4000 * 12.5 * (t + 2)]/100$$

$$\Rightarrow [4000 * 12.5 * (2 + 2)]/100$$

$$\Rightarrow \text{Rs. 2000}$$

12. Questions

Since, the ratio of the rate of interest in scheme A to that in scheme B = 3: 5

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So, let the rates of interest in schemes A and B be '3r'% and '5r'% respectively.

Since, the difference between compound interest and simple interest on Rs. 8000, which is invested in scheme A, after 2 years is Rs. X.

So,

$$\Rightarrow 8000 * (3r/100)^2 = X \text{ -----(1)}$$

Since, the difference between compound interest and simple interest on Rs. 4000, which is invested in scheme B, after 2 years is Rs. (X + 17.5).

So,

$$\Rightarrow 4000 * (5r/100)^2 = (X + 17.5) \text{ -----(2)}$$

By equation (2) – (1):

$$\Rightarrow 4000 * (5r/100)^2 - 8000 * (3r/100)^2 = X + 17.5 - X$$

$$\Rightarrow 4000 * (25r^2/10000) - 8000 * (9r^2/10000) = 17.5$$

$$\Rightarrow 4 * (25r^2/10) - 8 * (9r^2/10) = 17.5$$

$$\Rightarrow 100r^2 - 72r^2 = 175$$

$$\Rightarrow r^2 = 6.25$$

$$\Rightarrow r = 2.5$$

The rate of interest in scheme A = $3 * 2.5 = 7.5\%$

The rate of interest in scheme B = $5 * 2.5 = 12.5\%$

From equation (1):

$$8000 * (3r/100)^2 = X$$

$$8000 * (7.5/100)^2 = X$$

$$X = 45$$

Since, the rate of interest in scheme A is 93.75% of that in scheme C. @QuizByMickey

So, the rate of interest in scheme C = $7.5 * (100/93.75) = 8\%$

If an amount of Rs. 7500 is invested in scheme D, then the simple interest received from scheme D after 2 years will be Rs. 300 less than that of received from scheme A after 2 years.

Let the rate of interest in scheme D be R%.

Since, the simple interest received from scheme A after 2 years = $(8000 * 7.5 * 2)/100 = \text{Rs. } 1200$

So, the simple interest received by from scheme D after 2 years:

$$\Rightarrow (7500 * R * 2)/100 = 1200 - 300$$

$$\Rightarrow 150R = 900$$

$$\Rightarrow R = 6$$

So, the rate of interest in scheme D = 6%

Answer: E

Since, the original rate of interest in scheme B = 12.5%

So, the new rate of interest in scheme B would be = $(100 - 28)\%$ of 12.5 = 9%

From I:

The ratio of the rate of interest in scheme A to that in scheme B = 7.5: 9 = 5: 6

So, I is true.

From II:

The SI received from scheme B on Rs. 3500 after 2 years:

$$\Rightarrow (3500 * 2 * 9)/100$$

$$\Rightarrow \text{Rs. } 630$$

So, II is true.

From III:

Since, the new rate of interest in scheme B is 9%, which is not a prime number.

So, III is not true.

Hence, only I and II are true.

13. Questions

Since, the ratio of the rate of interest in scheme A to that in scheme B = 3: 5

So, let the rates of interest in schemes A and B be '3r%' and '5r%' respectively.

Since, the difference between compound interest and simple interest on Rs. 8000, which is invested in scheme A, after 2 years is Rs. X.

So,

$$\Rightarrow 8000 * (3r/100)^2 = X \text{ -----(1)}$$

Since, the difference between compound interest and simple interest on Rs. 4000, which is invested in scheme B, after 2 years is Rs. (X + 17.5).

So,

$$\Rightarrow 4000 * (5r/100)^2 = (X + 17.5) \text{ -----(2)}$$

By equation (2) – (1):

$$\Rightarrow 4000 * (5r/100)^2 - 8000 * (3r/100)^2 = X + 17.5 - X$$

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$$\Rightarrow 4000 * (25r^2/10000) - 8000 * (9r^2/10000) = 17.5$$

$$\Rightarrow 4 * (25r^2/10) - 8 * (9r^2/10) = 17.5$$

$$\Rightarrow 100r^2 - 72r^2 = 175$$

$$\Rightarrow r^2 = 6.25$$

$$\Rightarrow r = 2.5$$

The rate of interest in scheme A = $3 * 2.5 = 7.5\%$

The rate of interest in scheme B = $5 * 2.5 = 12.5\%$

From equation (1):

$$8000 * (3r/100)^2 = X$$

$$8000 * (7.5/100)^2 = X$$

$$X = 45$$

Since, the rate of interest in scheme A is 93.75% of that in scheme C.

So, the rate of interest in scheme C = $7.5 * (100/93.75) = 8\%$

If an amount of Rs. 7500 is invested in scheme D, then the simple interest received from scheme D after 2 years will be Rs. 300 less than that of received from scheme A after 2 years.

Let the rate of interest in scheme D be R%.

Since, the simple interest received from scheme A after 2 years = $(8000 * 7.5 * 2)/100 = \text{Rs. } 1200$

So, the simple interest received by from scheme D after 2 years:

$$\Rightarrow (7500 * R * 2)/100 = 1200 - 300$$

$$\Rightarrow 150R = 900$$

$$\Rightarrow R = 6$$

So, the rate of interest in scheme D = 6%

Answer: E

Since, the SI received from scheme C on Rs. (P + 500) after 20 months is Rs. 1400.

And the rate of interest in scheme C = 8%

So,

$$[(P + 500) * 8 * 20]/(100 * 12) = 1400$$

$$(P + 500) * 160 = 1680000$$

$$P + 500 = 10500$$

$$P = 10000$$

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Since, the rate of interest in scheme D = 6%

So, the SI received from scheme D on Rs. (P – 2500) after 15 months:

$$\Rightarrow [(P - 2500) * 6 * 15] / (100 * 12)$$

$$\Rightarrow [(10000 - 2500) * 6 * 5] / (100 * 4)$$

$$\Rightarrow [7500 * 6 * 5] / (100 * 4)$$

$$\Rightarrow \text{Rs. } 562.5$$

Since, the SI received from scheme D lie between 548 and 568 and also between 555 and 575.

So, both options (b) and (d) are correct.

14. Questions

Answer: C

According to question,

$$19.99^2 - 12.99^2 + (4832.05 - x) / 19.88 = 500$$

$$20^2 - 13^2 + (4832 - x) / 20 = 500$$

$$(4832 - x) = 269 \times 20$$

$$\text{Thus, } x = -548$$

Hence, answer is option C

15. Questions

Answer: A

According to question,

$$\sqrt{1295.88 \times 49.90 + 1079.60 \div 24.05 - 44.05 \times 7.99} = x$$

$$\sqrt{1296 \times 50 + 1080 \div 24 - 44 \times 8} = x$$

$$x = 1493$$

Hence, answer is option A

16. Questions

Answer: B

According to question,

$$x = \sqrt{2208.99 \times 12.05 + 59.98 - 31.52\% \text{ of } 449.99}$$

$$x = \sqrt{2209 \times 12 + 60 - 144}$$

$$\text{Thus, } x = 480$$

Hence, answer is option B

17. Questions

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Answer: D

According to question,

$$660\sqrt{x} = (23.99 \times 14.98) - 227.98$$

$$660\sqrt{x} = (24 \times 15) - 228$$

$$\sqrt{x} = 132/660$$

$$\sqrt{x} = 1/5$$

$$\text{Thus, } x = 1/25$$

Hence, answer is option D

18. Questions

Answer: B

Let the speeds of trains A and B be 'a' m/s and 'b' m/s respectively.

Since, trains A and B cross each other in 15 seconds while running in opposite direction.

So,

$$(a + b) = (235 + 350)/15$$

$$a + b = 39 \text{ -----(1)}$$

From statement I alone:

Since, train B crosses a $(d - 130)$ m long tunnel in 25 seconds.

So,

$$b = [(d - 130) + 350]/25$$

Since, the equation can't be solved, so the required time can't be determined.

So, statement I alone is not sufficient to answer the question.

From statements I and III together:

Since, the time taken by train A to cross a $(d + 110)$ m long bridge is 25 seconds.

So,

$$a = [(d + 110) + 235]/25 \text{ -----(2)}$$

From statement I:

$$b = [(d - 130) + 350]/25 \text{ -----(3)}$$

From equations (1), (2) and (3):

$$[(d + 110) + 235]/25 + [(d - 130) + 350]/25 = 39$$

$$d + 345 + d + 220 = 975$$

$$d = 205$$

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From equation (2):

$$a = [(d + 110) + 235]/25$$

$$a = [(205 + 110) + 235]/25$$

$$a = 22$$

Now, the time taken by train A to cross a 'd' m long platform = $(235 + 205)/22 = 20$ seconds

So, statements I and III together are sufficient to answer the question.

From statement II alone:

The time taken by train A to cross a man in train B is 47 seconds, while running in same direction.

Since, the man is travelling in train B, so the speed of the man will also be 'b' m/s.

So,

$$(a - b) = 235/47$$

$$a - b = 5 \text{ -----(4)}$$

From equations (1) and (4):

$$a = (39 + 5)/2$$

$$a = 22$$

Since, the value of 'd' can't be determined, so the required time can't be determined.

So, statement II alone is not sufficient to answer the question.

Hence, statements I and III together are sufficient to answer the question.

19. Questions

Answer: A

Given: $(M + 3) \times (N + 2) \times (O - 1) = 385$

385 can be written as: $1 \times 5 \times 7 \times 11$.

Since all M, N, and O are greater than 2.

Which means $(O - 1) \neq 1$ because it will make $O = 2$.

Also, $(M + 3) \neq 5$ because it will make $M = 2$

So, values of $(M + 3)$, $(N + 2)$, $(O - 1)$ can be either 5 or 7 or 11 (Not in the same order).

Case 1: When $(M + 3) = 7$, $(N + 2) = 5$, and $(O - 1) = 11$

$$M = 4, N = 3, \text{ and } O = 12$$

$$M + N - O = 4 + 3 - 12 = -5$$

Invalid case

Case 2: When $(M + 3) = 7$, $(N + 2) = 11$, and $(O - 1) = 5$

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$M = 4$, $N = 9$, and $O = 6$

$$M + N - O = 4 + 9 - 6 = 7$$

Invalid case

Case 3: When $(M + 3) = 11$, $(N + 2) = 5$, and $(O - 1) = 7$

$M = 8$, $N = 3$, and $O = 8$

$$M + N - O = 8 + 3 - 8 = 3$$

Valid case

Case 4: When $(M + 3) = 11$, $(N + 2) = 7$, and $(O - 1) = 5$

$M = 8$, $N = 5$, and $O = 6$

$$M + N - O = 8 + 5 - 6 = 7$$

Invalid case

From statement I alone:

Value of $M + N - O$ in case 3 $= 8 + 3 - 8 = 3$ (Lies between 2 and 5.)

$M = O = 8$ [So M is not largest]

Statement I alone is not sufficient.

From statement II alone:

Integer values of M and O are equal.

Only case 3 is possible.

$$\text{Value of } M + N - O = 8 + 3 - 8 = 3$$

Value of $M + N - O$ lies between 2 and 5.

Statement II alone is sufficient.

From statement III alone:

Sum of integer values of N and O is a prime number and M^O is a perfect square integer.

Sum of values of N and O in case 3 $= 3 + 8 = 11$ [A prime number.]

Now,

M^O in case 3 $= 8^8$ (A perfect square.)

Statement III alone is sufficient.

20. Questions

Answer: D

Per day salary of the person in a month which have 30 days $= 30000/30 = \text{Rs. } 1000$

From I and II together:

Total salary for the month after 'x' half pay leaves and 'y' no pay leaves = $30000 - x \times 500 - y \times 1000 = 25500$

$$500x + 1000y = 4500$$

$$x + 2y = 9 \dots\dots\dots (1)$$

Deduction in salary due to half pay leaves = $x \times 500 = 500x$

Deduction in salary due to no pay leaves = $y \times 1000 = 1000y$

According to the question:

$$500x - 1000y = 500$$

$$x - 2y = 1 \dots\dots\dots (2)$$

From (1) and (2):

$$x = 5 \text{ and } y = 2$$

Total deduction in salary due to half pay leaves only = $500x = \text{Rs. } 2500$

So, statements I and II together are sufficient.

From I and III together:

Total salary for the month after 'x' half pay leaves and 'y' no pay leaves = $30000 - x \times 500 - y \times 1000 = 25500$

$$500x + 1000y = 4500$$

$$x + 2y = 9 \dots\dots\dots (1)$$

Total deduction in his salary due to half pay leaves = $x \times 500 = 500x$

Total deduction in his salary due to no pay leaves = $y \times 1000 = 1000y$

According to the question:

$$500x + 1000y = 4500$$

$$x + 2y = 9 \dots\dots\dots (2)$$

Equations (1) and (2) are same. So, we cannot calculate the value of 'x' and 'y'.

So, statements I and III together are not sufficient.

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From II and III together:

Deduction in salary due to half pay leaves = $x \times 500 = 500x$

Deduction in salary due to no pay leaves = $y \times 1000 = 1000y$

According to the question:

$$500x - 1000y = 500$$

$$x - 2y = 1 \dots\dots\dots (1)$$

Total deduction in his salary due to half pay leaves = $x \times 500 = 500x$

Total deduction in his salary due to no pay leaves = $y \times 1000 = 1000y$

According to the question:

$$500x + 1000y = 4500$$

$$x + 2y = 9 \dots\dots\dots (2)$$

From (1) and (2):

$$x = 5 \text{ and } y = 2$$

Total deduction in salary due to half pay leaves only = $500x = \text{Rs. } 2500$

So, statements II and III together are sufficient.

Hence, either statements I and II together are sufficient or statements II and III together are sufficient to answer the question.

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